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ELECTRICAL CONNECTION APPARATUS

Abstract

An electrical connection apparatus includes: a probe; a probe head holding the probe; a wiring sheet with flexibility including a first connection portion arranged on a first sheet surface facing the probe head and a second connection portion arranged on a second sheet surface; and a wiring board which is arranged facing the probe head with the wiring sheet therebetween and has a first electrode facing the second connection portion. The wiring sheet include: a first layer through which the first connection portion passes; a wiring layer which includes an electrode pad arranged facing the first connection portion and an internal circuit electrically connected to the electrode pad; and a second layer through which the second connection portion passes. The first layer is elastically deformed such that the first connection portion, which has received a pressing force from the probe, comes into contact with the electrode pad.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is based on, and claims priority from Japanese Patent Application No. 2024-029921, filed on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an electrical connection apparatus used for inspecting electrical characteristics of an object to be inspected.

BACKGROUND ART

[0003] An electrical connection apparatus having a probe is used to inspect electrical characteristics of an object to be inspected such as a semiconductor integrated circuit in a wafer state. In the inspection using the probe, one end of the probe contacts an electrode of the object to be inspected, and the other end of the probe contacts a terminal arranged on a wiring board in the electrical connection apparatus. The terminal arranged on the wiring board is electrically connected to an inspection device such as a tester.

CITATION LIST

Patent Literature

[0004] [Patent Literature 1] JP 2019-178901 A

SUMMARY

[0005] In order to accurately inspect the electrical characteristics of the object to be inspected, it is necessary to stably connect the probe to the object to be inspected. Therefore, it is necessary to press the probe against the wiring board. This applies a load to the probe. However, the probe has the load endurance limit. Therefore, an object of the present disclosure is to provide an electrical connection apparatus which enables reduction of a load applied to a probe.

[0006] An electrical connection apparatus according to one aspect of the present disclosure includes: a probe; a probe head holding the probe; a wiring sheet with flexibility including a first connection portion arranged on a first sheet surface facing the probe head and a second connection portion arranged on a second sheet surface; and a wiring board which is arranged facing the probe head with the wiring sheet therebetween and on which a first electrode facing the second connection portion is arranged. The wiring sheet includes: a first layer through which the first connection portion passes; a wiring layer which includes an electrode pad arranged on a first connection surface separated from the first layer so as to face the first connection portion, and an internal circuit electrically connected to the electrode pad; and a second layer through which the second connection portion **32** passes. The first layer is elastically deformed such that the first connection portion, which has received a pressing force from the probe, comes into contact with the electrode pad.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a schematic view showing a configuration of an electrical connection apparatus according to an embodiment.

[0008] FIG. **2A** is a schematic plan view showing the configuration of the electrical connection apparatus according to the embodiment.

[0009] FIG. 2B is a schematic cross-sectional view showing the configuration of the electrical connection apparatus according to the embodiment.

[0010] FIG. 3 is a schematic view showing the elastic deformation of a first layer of a wiring sheet in the electrical connection apparatus according to the embodiment.

[0011] FIG. 4 is a schematic view showing an original shape of the first layer of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0012] FIG. 5 is a schematic view showing the elastic deformation of a second layer of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0013] FIG. 6 is a schematic view showing an original shape of the second layer of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0014] FIG. 7 is a schematic view showing the shape of a first space formed in the first layer of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0015] FIG. 8 is a schematic view showing the shape of a second space formed in the second layer of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0016] FIG. 9 is a schematic cross-sectional view showing a configuration example of a wiring layer of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0017] FIG. 10 is a schematic view showing an example of an internal circuit of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0018] FIG. 11 is a schematic view showing another example of an internal circuit of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0019] FIG. 12 is a schematic view showing another example of an internal circuit of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0020] FIG. 13 is a schematic view showing an example of a loopback circuit using the wiring sheet shown in FIG. 12.

[0021] FIG. 14 is a schematic view showing another example of an internal circuit of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0022] FIG. 15 is a schematic view showing another example of an internal circuit of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0023] FIG. 16 is a schematic view showing an example of an arrangement of electronic circuits on the wiring sheet in the electrical connection apparatus according to the embodiment.

DETAILED DESCRIPTION

[0024] An embodiment will be described with reference to the drawings. In the description of the drawings below, the same or similar parts are denoted with the same or similar reference numerals. However, it should be noted that the drawings are schematically shown and the ratios of the thickness of each portion and the like are different from those in reality. Further, it is needless to say that the drawings include portions where the relationships and ratios of dimensions are different between drawings. The following embodiment exemplifies an apparatus and a method for embodying the technical concept of the present invention, and the embodiment does not specify the material, shape, structure, arrangement, and the like of components to the following.

[0025] An electrical connection apparatus **1** according to an embodiment shown in FIG. 1 is used for inspecting an object **2** to be inspected. The electrical connection apparatus **1** includes probes **10**, a probe head **60** holding the probes **10**, a wiring sheet **30** laminated on the probe head **60**, and a wiring board **20** laminated on the wiring sheet **30**.

[0026] The electrical connection apparatus **1** further includes a printed board **40** laminated on the wiring board **20** and a stiffener **50** laminated on the printed board **40**. In the following description, a direction in which the electrical connection apparatus **1** is located when viewed from the object **2** to be inspected is defined as an upward direction, and a direction in which the object **2** to be inspected is located when viewed from the electrical connection apparatus **1** is defined as a downward direction. As shown in FIG. 1, a direction from the downward direction to the upward direction is defined as a Z direction, and a plane perpendicular to the Z direction is defined as an

XY plane. In FIG. 1, a left-right direction of the drawing is defined as an X direction, and a front-back direction is defined as a Y direction. Further, a surface facing the upward direction of each component of the electrical connection apparatus **1** will be referred to as an upper surface, and a surface facing the downward direction will be referred to as a lower surface.

[0027] Each probe **10** has a tip **11** which is one end arranged so as to be in contact with the object **2** to be inspected and a proximal end **12** which is the other end connected to the tip **11**. The probes **10** are held while passing through through-holes formed in the probe head **60**, for example. Each tip **11** and each proximal end **12** are exposed from the probe head **60**.

[0028] The object **2** to be inspected is mounted on a stage **3**. The electrical connection apparatus **1** and the stage **3** are movable relative to one another in an up-down direction. When the object **2** to be inspected is inspected, a distance between the electrical connection apparatus **1** and the stage **3** is made small, and tips **11** of the probes **10** contact signal terminals (not shown) of the object **2** to be inspected. FIG. 1 shows a state in which the probes **10** and the object **2** to be inspected are separated.

[0029] The wiring sheet **30** includes first connection portions **31** arranged on a first sheet surface **311** facing the probe head **60**, and second connection portions **32** arranged on a second sheet surface **312** facing in an opposite direction of the first sheet surface **311**. The wiring sheet **30** has an internal circuit (not shown in FIG. 1) electrically connected to at least any of the first connection portions **31** and the second connection portions **32**. Each first connection portion **31** is connected to a proximal end **12** of each probe **10** which faces the first sheet surface **311** and which is exposed from the probe head **60**. The wiring sheet **30** has flexibility and elastically deforms in a thickness direction. Details of the configuration of the wiring sheet **30** will be described later.

[0030] The wiring board **20** is arranged facing the probe head **60** with the wiring sheet **30** therebetween. First electrodes **21** electrically connected to the second connection portions **32** of the wiring sheet **30** are arranged on a first main surface **201** of the wiring board **20** facing the second sheet surface **312** of the wiring sheet **30**. Second electrodes **22** are arranged on a second main surface **202** of the wiring board **20** which faces the printed board **40** and faces in an opposite direction of the first main surface **201**. The first electrodes **21** and the second electrodes **22** are electrically connected via internal wiring **200**. A multilayer wiring board such as a Multi-Layer Organic (MLO) board or a Multi-Layer Ceramic (MLC) board may be used for the wiring board **20**, for example.

[0031] The printed board **40** is arranged facing the second main surface **202** of the wiring board **20**. The printed board **40** has first ends **41** arranged on a lower surface thereof facing the wiring board **20**, second ends **42** arranged on an upper surface thereof, and wiring patterns **400** for electrically connecting the first ends **41** and the second ends **42**. The second ends **42** are electrically connected to an inspection device (not shown), for example. As a result, an electrical signal propagates between the object **2** to be inspected and the inspection device via the electrical connection apparatus **1**.

[0032] The wiring board **20** may be a space transformer that transforms a distance between the proximal ends **12** of the probes **10** to a distance between the first ends **41** of the printed board **40**, when viewed from a direction normal to the second main surface **202** of the wiring board **20**, for example. Due to the wiring board **20** being the space transformer, it is possible to electrically connect the signal terminals arranged on the object **2** to be inspected to the second ends **42** of the printed board **40** which are arranged at a distance larger than a distance between the signal terminals. This facilitates the electrical connection between the wiring patterns **400** in the printed board **40** and the inspection device.

[0033] As shown in FIG. 1, the stiffener **50** may be laminated on the printed board **40**. The stiffener **50** has higher stiffness than the printed board **40**, and ensures the mechanical strength of the electrical connection apparatus **1** by preventing the printed board **40** from being bent. In addition, the stiffener **50** may be used as a support for fixing each component of the electrical connection

apparatus **1**. The stiffener **50** may be fixed to the printed board **40** using screws, for example.

[0034] FIG. **2A** is a plan view (hereinafter also referred to as “in plan view”) of a structure in which the wiring board **20**, the printed board **40**, and the stiffener **50** are laminated, as viewed from the Z direction. FIG. **2B** is a cross-sectional view of a side surface as viewed from the Y direction of an XY plane perpendicular to the Z direction. FIG. **2B** does not show the first electrodes **21** and the second electrodes **22** of the wiring board **20**, the first connection portions **31** and the second connection portions **32** of the wiring sheet **30**, and the wiring patterns **400** in the printed board **40**.

[0035] The stiffener **50** is arranged on an upper surface of the printed board **40** that is circular in plan view. As shown in FIG. **2A**, the stiffener **50** has a shape in which an outer circular ring and an inner rectangular ring are connected using spokes, for example. The wiring board **20** and the wiring sheet **30** are arranged in the vicinity of the center of a lower surface of the printed board **40**. The wiring board **20**, the printed board **40**, and the stiffener **50** are fixed with screws, for example. FIG. **2B** shows an example in which the wiring board **20** and the printed board **40** are joined using screws **70**. The wiring sheet **30** is not screwed, and is interposed and held between the probe head **60** and the wiring board **20**. The wiring sheet **30** has holes for inserting the screws **70**.

[0036] Details of a configuration of the wiring sheet **30** will be described below. As shown in FIG. **1**, the wiring sheet **30** includes a first layer **350**, a wiring layer **360**, and a second layer **370**. The first layer **350**, the wiring layer **360**, and the second layer **370** are arranged in this order in a thickness direction of the wiring sheet **30** (Z direction).

[0037] Both surfaces of the first layer **350** are defined by the first sheet surface **311** and a first facing surface **352**. The first connection portions **31** pass through the first layer **350** from the first sheet surface **311** to the first facing surface **352**.

[0038] Both surfaces of the wiring layer **360** are defined by a first connection surface **361** facing the first facing surface **352** and a second connection surface **362** facing in an opposite direction of the first connection surface **361**. The wiring layer **360** includes electrode pads **363** arranged on the first connection surface **361** separated from the first layer **350** so as to face the first connection portions **31**, and an internal circuit (not shown) electrically connected to the electrode pads **363**.

[0039] Both surfaces of the second layer **370** are defined by a second facing surface **371** facing the second connection surface **362** and the second sheet surface **312**. The second connection portions **32** pass through the second layer **370** from the second facing surface **371** to the second sheet surface **312**.

[0040] The peripheries of the first connection portions **31** of the first layer **350** are elastically deformed such that the first connection portions **31**, which have received pressing forces from the proximal ends **12** of the probes **10**, contact the electrode pads **363**.

[0041] The second layer **370** is separated from the second connection surface **362** in the peripheries of the second connection portions **32**. The peripheries of the second connection portions **32** of the second layer **370** are elastically deformed such that the second connection portions **32** are energized toward the first electrodes **21**.

[0042] When the electrical connection apparatus **1** is assembled, a pressing force is applied to press the first connection portions **31** against the proximal ends **12** of the probes **10**. Pressing the first connection portions **31** against the proximal ends **12** will be hereinafter referred to as “preloading”. The probe head **60** holding the probes **10** is attached to the wiring sheet **30**. Then, the first layer **350** is elastically deformed such that each first connection portion **31** contacts each electrode pad **363** as shown in FIG. **3**. In other words, the first layer **350** is elastically deformed during preloading. The first connection portions **31** of the wiring sheet **30** are pressed against the probes **10** by an elastic force of the first layer **350**. As a result, the probes **10** and the first connection portions **31** are in stable contact.

[0043] When the probes **10** are separated from the wiring sheet **30**, a shape of the first layer **350** returns to an original shape such that each first connection portion **31** is away from each electrode pad **363** as shown in FIG. **4**. In other words, FIG. **4** shows a state of the wiring sheet **30** before the

probe head **60** is attached to the wiring sheet **30**.

[0044] The peripheries of the second connection portions **32** of the second layer **370** are formed to be elastically deformable. Therefore, when the wiring board **20** is attached to the wiring sheet **30**, each second connection portion **32** of the wiring sheet **30** energized against each first electrode **21** of the wiring board **20** is pressed against the wiring layer **360** as shown in FIG. 5. Each second connection portion **32** and each first electrode **21** are reliably contacted by an elastic force of the elastically deformed second layer **370**. A gold ball or the like may be used for each second connection portion **32**, for example.

[0045] When the wiring board **20** is separated from the wiring sheet **30**, a shape of the periphery of each second connection portion **32** of the second layer **370** returns to an original shape as shown in FIG. 6. In other words, FIG. 6 shows a state of the wiring sheet **30** before the wiring sheet **30** is attached to the wiring board **20**.

[0046] As shown in FIGS. 3 and 4, first spacers **381** are arranged in regions in the periphery of a region where each first connection portion **31** is arranged, between the first layer **350** and the wiring layer **360** of the wiring sheet **30**. A space is formed between the first layer **350** and the wiring layer **360** by the first spacers **381**. Due to the first spacers **381** being arranged in the periphery of each first connection portion **31**, the first layer **350** is curved in the space range between the first layer **350** and the wiring layer **360**. In other words, the degree of curvature of the first layer **350** is limited by the thickness of each first spacer **381** in the Z direction.

[0047] As shown in FIGS. 5 and 6, second spacers **382** are arranged in regions in the periphery of a region where each second connection portion **32** is arranged, between the second layer **370** of the wiring sheet **30** and the wiring board **20**. A space is formed between the second layer **370** and the wiring board **20** by the second spacers **382**. Due to the second spacers **382** being arranged in the periphery of each second connection portion **32**, the second layer **370** is curved in the space range between the second layer **370** and the wiring board **20**. In other words, the degree of curvature of the second layer **370** is limited by the thickness of each second spacer **382** in the Z direction.

[0048] In order to elastically deform the periphery of each first connection portion **31** of the first layer **350**, first spaces **310** for bending may be disposed in the periphery of each first connection portion **31** in the first layer **350** as shown in FIG. 7, for example. The periphery of each first connection portion **31** can be elastically deformed by arranging each first connection portion **31** in a region with flexibility and with the first spaces **310** disposed in the periphery thereof. The magnitude of an elastic force can be adjusted by setting the area size of each first space **310**.

[0049] In order to elastically deform the periphery of each second connection portion **32** of the second layer **370**, second spaces **320** for bending may be disposed in the periphery of each second connection portion **32** in the second layer **370** as shown in FIG. 8, for example. The periphery of each second connection portion **32** can be elastically deformed by arranging each second connection portion **32** in a region with flexibility and with the second spaces **320** disposed in the periphery thereof. The magnitude of an elastic force can be adjusted by setting the area size of each second space **320**. The area of each first space **310** and the area of each second space **320** may be different. If an elastic force of the second layer **370** may be smaller than that of the first layer **350**, the area of each second space **320** may be larger than that of each first space **310**, for example.

[0050] Insulating films may be used for the first layer **350** and the second layer **370** of the wiring sheet **30**, for example. The wiring layer **360** may have a structure in which a conductive film and an insulating film are laminated. An internal circuit may be constituted from a conductive pattern formed on a conductive film, for example. FIG. 9 shows an example of a configuration of the wiring layer **360** of the wiring sheet **30**. FIG. 9 does not show the electrode pads **363** and the like.

[0051] The wiring layer **360** of the wiring sheet **30** shown in FIG. 9 has a structure in which a laminated body constituted by conductive films **302** and insulating films **303** is interposed between a pair of cover films **301** of insulating materials. The number of laminated bodies constituted by the conductive films **302** and the insulating films **303** can be set to any number. The conductive films

302 may be metal materials such as copper foils, for example. The insulating films **303** may be insulating materials such as polyimide sheets, for example. The cover films **301** may be insulating materials such as solder resists, for example. An adhesive may be used for joining the conductive films **302**, the insulating films **303**, and the cover films **301** each other, for example. The wiring layer **360** may have a structure in which a film (also referred to as a “base film”) of an insulating material is further interposed between laminated bodies including the conductive films **302** and the insulating films **303**. The cover films **301** may be used for the first layer **350** and the second layer **370**.

[0052] The wiring sheet **30** may be selected from among a plurality of wiring sheet candidates. Each of the wiring sheet candidates has a first connection portion **31** arranged on the first sheet surface **311**, a second connection portion **32** arranged on the second sheet surface **312**, and a wiring layer **360** on which an internal circuit is formed. Each of the wiring sheet candidates may include an internal circuit having a configuration different from that of another wiring sheet candidate. One wiring sheet **30** selected from among the plurality of wiring sheet candidates may be configured in an attachable/detachable manner between the probe head **60** and the wiring board **20**.

[0053] An example of the configuration of wiring sheet candidates included in a wiring sheet group will be described below. In the following, when each of the wiring sheet candidates is not limited, the wiring sheet candidates will be collectively referred to as a wiring sheet **30**.

[0054] An internal circuit of a wiring sheet **30** as any one of the group of wiring sheets may include a circuit for electrically connecting the first connection portions **31** and the second connection portions **32** (hereinafter also referred to as an “interposer circuit”). When the interposer circuit is formed on the wiring layer **360** having the structure shown in FIG. **9**, wiring is formed, which passes through the cover films **301**, the conductive films **302**, and the insulating films **303**, from the first connection surface **361** to the second connection surface **362** of the wiring layer **360**, for example.

[0055] If the internal circuit of the wiring sheet **30** includes the interposer circuit, the internal circuit of the wiring sheet **30** shown in FIG. **10** may include a circuit for short-circuiting the first connection portions **31** and the second connection portions **32**. In the internal circuit of the wiring sheet **30** shown in FIG. **10**, the first connection portions **31** and the second connection portions **32** are electrically short-circuited by short-circuit wiring **331**.

[0056] By attaching the wiring sheet **30** shown in FIG. **10** to the electrical connection apparatus **1**, the probes **10** and the wiring patterns **400** in the printed board **40** are short-circuited via the internal circuit of the wiring sheet **30**. This electrically connects the object **2** to be inspected and the inspection device. As a result, an electrical signal propagates between an inspection device such as an IC tester and the object **2** to be inspected, and characteristics of the object **2** to be inspected are measured. One first connection portion **31** and one second connection portion **32** are connected in a one-to-one relationship by means of the internal circuit of the wiring sheet **30**, for example. Alternatively, one first connection portion **31** may be connected to a plurality of second connection portions **32**, or a plurality of first connection portions **31** may be connected to one second connection portion **32** by means of the internal circuit.

[0057] When the internal circuit of the wiring sheet **30** includes the interposer circuit, the internal circuit may include a matching circuit **332** having a first terminal connected to a first connection portion **31** and a second terminal connected to a second connection portion **32** as shown in FIG. **11**. The matching circuit **332** may perform impedance matching between the first connection portion **31** and the second connection portion **32**. The matching circuit **332** may include a x-type filter, for example.

[0058] The internal circuit of the wiring sheet **30** may include a circuit for electrically connecting any one of the first connection portions **31** and another one of the first connection portions **31**. In other words, the internal circuit of the wiring sheet **30** may include a circuit for electrically connecting an output terminal and an input terminal of the object **2** to be inspected (hereinafter also

referred to as a “loopback circuit”). Two signal terminals of the object **2** to be inspected are electrically connected by the loopback circuit.

[0059] The internal circuit of the wiring sheet **30** may include a circuit for short-circuiting any one of the first connection portions **31** and another one of the first connection portions **31** as shown in FIG. **12**, for example. In the internal circuit of the wiring sheet **30** shown in FIG. **12**, the one first connection portion **31** and the other first connection portion **31** are electrically short-circuited by loopback wiring **333**. By attaching the wiring sheet **30** shown in FIG. **12** to the electrical connection apparatus **1**, one of the probes **10** and another one of the probes **10** are short-circuited via the internal circuit of the wiring sheet **30**. This electrically connects one signal terminal and the other signal terminal of the object **2** to be inspected.

[0060] FIG. **13** shows a configuration in which a first signal terminal **2A** of the object **2** to be inspected in contact with one of the probes **10**, and a second signal terminal **2B** of the object **2** to be inspected in contact with another one of the probes **10** are electrically connected via the loopback wiring **333** of the wiring sheet **30**. Suppose that the object **2** to be inspected is a receiving circuit, the first signal terminal **2A** is an output terminal of the object **2** to be inspected, and the second signal terminal **2B** is an input terminal of the object **2** to be inspected, for example. In the above case, a transmission test can be performed by returning an output from the object **2** to be inspected to an input. In other words, it is possible to test whether an output portion and an input portion of the object **2** to be inspected are functioning normally, even if there is no destination equipment for transmission. As an inspection in accordance with a jitter tolerance test performed for the receiving circuit, an output signal output from the first signal terminal **2A** (output terminal) may be input to the second signal terminal **2B** (input terminal) as an input signal to inspect whether a specified error rate is ensured, for example.

[0061] When the internal circuit of the wiring sheet **30** includes the loopback circuit, the internal circuit may include a circuit having a capacitor **34** connected in series between any one of the first connection portions **31** and another one of the first connection portions **31** as shown in FIG. **14**. One terminal of the capacitor **34** is connected to the one first connection portion **31**, and the other terminal of the capacitor **34** is connected to the other first connection portion **31**. The capacitor **34** may be a capacitor formed using a semiconductor manufacturing process (hereinafter also referred to as a “process capacitor”).

[0062] Further, the internal circuit of the wiring sheet **30** may include a relay circuit that switches to electrically connect one of the first connection portions **31** to either one of the second connection portions **32** or another one of the first connection portions **31**. The internal circuit shown in FIG. **15** includes a relay circuit **334** that constitutes either the interposer circuit for connecting the first connection portions **31** and the second connection portion **32** or the loopback circuit for connecting the first connection portions **31** each other, for example.

[0063] When the relay circuit **334** shown in FIG. **15** constitutes the interposer circuit, a first contact terminal **334a** and a second contact terminal **334b** are connected, and a third contact terminal **334c** and a fourth contact terminal **334d** are connected. This electrically connects the first connection portions **31** and the second connection portions **32**. When the relay circuit **334** constitutes the loopback circuit, the first contact terminal **334a** and a common contact terminal **334e** of the relay circuit **334** are connected, and the third contact terminal **334c** and the common contact terminal **334e** are connected. This electrically connects one of the first connection portions **31** and another one of the first connection portions **31**.

[0064] Although examples of the internal circuit of the wiring sheet **30** have been described with reference to FIGS. **10** to **15**, it is needless to say that the configuration of the internal circuit is not limited to the above. The internal circuit may include an inductor instead of the capacitor **34** shown in FIG. **14**, or the internal circuit may include both a capacitor and an inductor, for example. In other words, the internal circuit of the wiring sheet **30** may include a passive circuit including any element. Further, the internal circuit may include a switching circuit using a diode or the like

instead of the relay circuit **334** shown in FIG. **15**.

[0065] An element included in the internal circuit of the wiring sheet **30** may be formed using a Micro Electro Mechanical Systems (MEMS) process, for example. By using the MEMS process, an element which is reduced in size can be formed integrally with the wiring sheet **30**.

[0066] As described above, various circuit configurations as shown in FIGS. **10** to **15** can be implemented in the electrical connection apparatus **1** by merely replacing the wiring sheet **30** in the electrical connection apparatus **1**, for example. Therefore, a plurality of types of measurements can be performed on the object **2** to be inspected using the electrical connection apparatus **1**. A DC test may be performed on the object **2** to be inspected by mounting, in the electrical connection apparatus **1**, the wiring sheet **30** including the internal circuit for short-circuiting the first connection portions **31** and the second connection portions **32**, for example. Further, a high-frequency test may be performed on the object **2** to be inspected by mounting, in the electrical connection apparatus **1**, the wiring sheet **30** including the internal circuit having the matching circuit or the loopback circuit.

[0067] Further, by arranging the relay circuit **334** in the wiring sheet **30** shown in FIG. **15** of the electrical connection apparatus **1**, the wiring length of the interposer circuit and the loopback circuit can be reduced, compared to that when a relay element is arranged on the printed board **40**. As a result, according to the electrical connection apparatus **1**, it is possible to shorten a propagation path of an electrical signal and suppress the loss of an electrical signal and noise.

[0068] Further, by arranging the matching circuit **332** in the wiring sheet **30** of the electrical connection apparatus **1** as shown in FIG. **11**, wiring connected to the matching circuit **332** can be shortened. The matching circuit **332** can be arranged in the immediate vicinity of wiring for which impedance matching is to be performed, and therefore impedance matching can be performed effectively, for example.

[0069] As described above, the electrical connection apparatus **1** has the wiring sheet **30** including the internal circuit capable of constituting any circuit, which is interposed between the wiring board **20** and the printed board **40** in an attachable/detachable manner, and therefore it is possible to configure an arbitrary measurement system. As a result, characteristics of the object **2** to be inspected can be measured with high accuracy with the electrical connection apparatus **1**.

[0070] Further, the wiring sheet **30** with flexibility which is interposed between the probe head **60** and the wiring board **20** in the electrical connection apparatus **1** can mitigate warpages and irregularities of surfaces of the probe head **60** and the wiring board **20**. This can suppress the rattling and contact failure of the electrical connection apparatus caused by a gap between the probe head **60** and the wiring board **20**.

[0071] Further, electronic components electrically connected to the internal circuit can be arranged on a surface of the wiring sheet **30** in the electrical connection apparatus **1**. As shown in FIG. **16**, electronic components **100** connected to the internal circuit of the wiring sheet **30** may be arranged on both of the first sheet surface **311** and the second sheet surface **312**, for example. Each electronic component **100** may be a capacitor, an inductor, or a resistance element, for example. FIG. **16** shows an example of arranging the electronic components **100** on both of the first sheet surface **311** and the second sheet surface **312**, but the electronic components **100** may be arranged on either the first sheet surface **311** or the second sheet surface **312**. By arranging the electronic components **100** on a surface of the wiring sheet **30**, it is possible to reduce or divide the number of electronic components arranged on the wiring board **20** or the printed board **40**, for example. Further, by arranging electronic components on the surface of the wiring sheet **30**, it is possible to reduce the wiring length between the object **2** to be inspected and each electronic component. Therefore, by arranging electronic components such as capacitors for reducing power source noise on the wiring sheet **30**, it is possible to stably perform measurements on the object **2** to be inspected, for example.

[0072] As described above, during preloading in an assembly process of the electrical connection

apparatus **1**, after the proximal ends **12** of the probes **10** and the first connection portions **31** of the first layer **350** of the wiring sheet **30** come into contact with each other, the first layer **350** is pushed toward the wiring layer **360** by the probes **10**. The first layer **350** is curved, and the degree of curvature is limited by the thickness of each first spacer **381**. Then, the first connection portions **31** come into contact with the electrode pads **363**. The proximal ends **12** of the probes **10** are electrically connected to the internal wiring **200** in the wiring board **20** via the internal circuit of the wiring sheet **30**.

[0073] Therefore, according to the electrical connection apparatus **1**, the curvature of the first layer **350** mitigates a part of a load applied until the first connection portions **31** of the first layer **350** come into contact with the electrode pads **363** of the wiring layer **360**. This can reduce the load applied to the probes **10** during preloading. As a result, breakage of the probes **10** can be prevented.

[0074] Meanwhile, all loads due to preloading are applied to probes **10** of an electrical connection apparatus of a comparative example not including the wiring sheet **30**. Therefore, it is necessary to reduce the loads applied to the probes **10** of the electrical connection apparatus of the comparative example, compared with those in the electrical connection apparatus **1**.

[0075] As described above, according to the electrical connection apparatus **1**, the load applied to the probes **10** can be reduced. This can increase the maximum load applied to the probes **10** when the object **2** to be inspected is inspected. Therefore, it is possible to prevent breakage of the probes **10** caused by bringing the electrical connection apparatus **1** close to the object **2** to be inspected in order to ensure the electrical connection between the probes **10** and the object **2** to be inspected. When positions of the tips **11** of the probes **10** vary, the electrical connection apparatus **1** can be brought close to the object **2** to be inspected such that a probe **10** having a long distance from the object **2** to be inspected reliably contacts the object **2** to be inspected, for example. At this time, the breakage of a probe **10** having a short distance from the object **2** to be inspected can be prevented due to the maximum load applied to the probes **10** being large.

[0076] The limit of the load applied to the probes **10** increases according to the length of the probes **10**. Therefore, according to the electrical connection apparatus **1** which enables reduction of the load applied to the probes **10**, the length of the probes **10** of the electrical connection apparatus **1** can be reduced compared to that of probes of the electrical connection apparatus of the comparative example. By reducing the length of the probes **10**, signal wiring in the electrical connection apparatus **1** can be shortened. Therefore, according to the electrical connection apparatus **1**, it is possible to enhance the accuracy of high-frequency current characteristics of the object **2** to be inspected.

OTHER EMBODIMENTS

[0077] Although an embodiment of the present invention has been described above, the discussion and drawings forming part of this disclosure should not be construed as limiting the invention. Various alternative embodiments, examples, and operational techniques will be apparent to those skilled in the art from this disclosure.

[0078] The wiring sheet **30** may include an internal circuit in which a plurality of types of circuit configurations are mixed, for example. The wiring sheet **30** may include an internal circuit in which the short-circuit wiring **331** and the matching circuit **332** are mixed, for example.

Alternatively, the wiring sheet **30** may include an internal circuit in which an interposer circuit and a loopback circuit are mixed. The internal circuit may include the short-circuit wiring **331**, the matching circuit **332**, and the loopback wiring **333**, or may further include the relay circuit **334**, for example. Further, the loopback circuit may include the matching circuit **332**. In this way, any circuit can be configured in the internal circuit of the wiring sheet **30**.

[0079] As described above, by mounting, in the electrical connection apparatus **1**, the wiring sheet **30** including the internal circuit in which any circuit configurations are mixed, a plurality of types of measurements can be performed on the object **2** to be inspected with one wiring sheet **30**.

[0080] In this way, it is needless to say that the present invention includes various embodiments not

described above. Therefore, the technical scope of the present invention is defined only by matters specified in the invention that are within the scope of claims appropriate from the above description.

REFERENCE SIGNS LIST

[0081] **1** Electrical connection apparatus [0082] **2** Object to be inspected [0083] **10** Probe [0084] **11** Tip [0085] **12** Proximal end [0086] **20** Wiring board [0087] **21** First electrode [0088] **22** Second electrode [0089] **30** Wiring sheet [0090] **31** First connection portion [0091] **32** Second connection portion [0092] **40** Printed board [0093] **41** First end [0094] **42** Second end [0095] **50** Stiffener [0096] **60** Probe head [0097] **100** Electronic component [0098] **200** Internal wiring [0099] **201** First main surface [0100] **202** Second main surface [0101] **311** First sheet surface [0102] **312** Second sheet surface [0103] **350** First layer [0104] **352** First facing surface [0105] **360** Wiring layer [0106] **361** First connection surface [0107] **362** Second connection surface [0108] **363** Electrode pad [0109] **370** Second layer [0110] **371** Second facing surface [0111] **381** First spacer [0112] **382** Second spacer [0113] **400** Wiring pattern

Claims

1. An electrical connection apparatus used for inspecting an object to be inspected, the electrical connection apparatus comprising: a probe having a tip arranged so as to be in contact with the object to be inspected and a proximal end connected to the tip; a probe head holding the probe; a wiring sheet with flexibility including a first connection portion arranged on a first sheet surface facing the probe head and a second connection portion arranged on a second sheet surface facing in an opposite direction of the first sheet surface; and a wiring board which is arranged facing the probe head with the wiring sheet therebetween and has a first main surface on which a first electrode facing the second connection portion is arranged, wherein the wiring sheet includes: a first layer which has both surfaces defined by the first sheet surface and a first facing surface and through which the first connection portion passes from the first sheet surface to the first facing surface; a wiring layer which has both surfaces defined by a first connection surface facing the first facing surface and a second connection surface facing in an opposite direction of the first connection surface, and which includes an electrode pad arranged on the first connection surface separated from the first layer so as to face the first connection portion, and an internal circuit electrically connected to the electrode pad; and a second layer which has both surfaces defined by a second facing surface facing the second connection surface and the second sheet surface and through which the second connection portion passes from the second facing surface to the second sheet surface, and the first layer is elastically deformed such that the first connection portion, which has received a pressing force from the proximal end, comes into contact with the electrode pad.
2. The electrical connection apparatus according to claim 1, wherein the second layer is separated from the second connection surface at the periphery of the second connection portion, and the periphery of the second connection portion of the second layer is elastically deformed such that the second connection portion is energized toward the first electrode.
3. The electrical connection apparatus according to claim 1, wherein a first spacer constituting a space formed between the first layer and the wiring layer is arranged in the periphery of the first connection portion, between the first layer and the wiring layer.
4. The electrical connection apparatus according to claim 1, wherein a second spacer constituting a space formed between the second layer and the wiring board is arranged in the periphery of the second connection portion, between the second layer and the wiring board.
5. The electrical connection apparatus according to claim 1, wherein the wiring sheet has a structure in which a laminated body constituted by a conductive film and an insulating film is interposed between cover films of an insulating material.
6. The electrical connection apparatus according to claim 1, further comprising: a printed board that

is arranged facing the wiring board and has a wiring pattern connected to a second electrode arranged on a second main surface facing in an opposite direction of the first main surface of the wiring board.

7. The electrical connection apparatus according to claim 6, wherein the probe is provided in plurality, the wiring pattern is provided in plurality, and the wiring board is a space transformer which transforms a distance between the probes into a distance between the wiring patterns of the printed board, when viewed from a direction normal to the main surface of the wiring board.
